Decision-Making Under Uncertainty: Exam Solutions

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Learning objective

Apply minimax regret and Expected Monetary Value (EMV) to select an alternative under uncertainty, and use sensitivity analysis to determine how a subjective probability changes the preferred alternative.

Criteria assessment

Each problem is worth 2 points. Show all calculations, including the complete regret table for C5, every EMV calculation for C6, and the threshold equation and decision ranges for C7. Partial credit may be awarded when the correct procedure is clearly shown but arithmetic errors are made. No points will be awarded if no procedure is shown. A **minimum of 4 points** is required to pass each criterion. A 1-point penalty will be applied to each criterion if the student's name is not written on the assessment or if the assessment is submitted late.

Question	**C5**	**C6**	**C7**
1	2	–	–
2	2	–	–
3	2	–	–
4	–	2	–
5	–	2	–
6	–	2	–
7	–	–	2
8	–	–	2
9	–	–	2
Total	**6**	**6**	**6**

1. Artisan Apiary Expansion [C5-2]

An artisan apiary must choose one of three hive-expansion plans. Annual net benefits are in thousands of dollars.

	Alternatives		
	Large	Medium	Small
Excellent	76	58	38
Average	34	49	43
Poor	-14	23	35

What decision would be selected using the Minimax Regret criterion? Justify your answer.

C5 — Solution

Here A_i follows the order of alternatives in the exam: Large, Medium, Small. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly.

The best payoff in each state is

$$b(S_1) = \max\{76, 58, 38\} = 76, \quad b(S_2) = \max\{34, 49, 43\} = 49, \quad b(S_3) = \max\{-14, 23, 35\} = 35.$$

Using $r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j)$ gives

	S_1	S_2	S_3	Maximum regret
A_1	$76 - 76 = 0$	$49 - 34 = 15$	$35 - (-14) = 49$	49
A_2	$76 - 58 = 18$	$49 - 49 = 0$	$35 - 23 = 12$	18
A_3	$76 - 38 = 38$	$49 - 43 = 6$	$35 - 35 = 0$	38

Thus $\min\{49, 18, 38\} = 18$, so the Minimax Regret criterion selects A_2, the medium expansion.

2. Regional Theater Capacity **[C5-2]**

A theater operator is choosing a seating-capacity plan. Seasonal profit is in thousands of dollars; no reliable probabilities are available.

Alternatives	High	Normal	Low
Add two seating sections	108	40	-28
Add one seating section	82	61	14
Lease a nearby hall	59	52	34
Current seating	34	34	34

What decision would be selected using the Minimax Regret criterion? Justify your answer.

C5 — Solution

Here A_i follows the order of alternatives in the exam: add two seating sections, add one seating section, lease a nearby hall, current seating. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly.

The statewise best payoffs are $b(S_1) = 108$, $b(S_2) = 61$, and $b(S_3) = 34$.

	S_1	S_2	S_3	Maximum regret
A_1	$108 - 108 = 0$	$61 - 40 = 21$	$34 - (-28) = 62$	62
A_2	$108 - 82 = 26$	$61 - 61 = 0$	$34 - 14 = 20$	26
A_3	$108 - 59 = 49$	$61 - 52 = 9$	$34 - 34 = 0$	49
A_4	$108 - 34 = 74$	$61 - 34 = 27$	$34 - 34 = 0$	74

Since $\min\{62, 26, 49, 74\} = 26$, the recommended decision is A_2 : add one seating section.

3. Emergency Shelter Staffing **[C5-2]**

A relief organization must select an emergency-shelter staffing plan. Net community value is in thousands of dollars.

	Alternatives		
	Full	Flexible	Limited
Very high	96	74	48
High	54	64	51
Moderate	24	49	46
Low	-24	12	38

What decision would be selected using the Minimax Regret criterion? Justify your answer.

C5 — Solution

Here A_i follows the order of alternatives in the exam: Full, Flexible, Limited. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly.

The best-payoff vector is

$$(b(S_1), b(S_2), b(S_3), b(S_4)) = (96, 64, 49, 38).$$

	S_1	S_2	S_3	S_4	Maximum regret
A_1	0	10	25	62	62
A_2	22	0	0	26	26
A_3	48	13	3	0	48

For example, $r(A_1, S_4) = 38 - (-24) = 62$ and $r(A_3, S_2) = 64 - 51 = 13$. The smallest maximum regret is $\min\{62, 26, 48\} = 26$, so select A_2, the flexible staffing plan.

4. Coffee Roastery Product Mix **[C6-2]**

A coffee roastery is selecting a product mix. Profits are in thousands of dollars.

		Strong	Stable	Weak
A_i	$P(S_j)$	0.45	0.35	0.20
Single-origin		86	44	-12
Balanced		64	58	22
Classic		42	47	37

What decision would be selected using the EMV or Bayes criterion? Justify your answer.

C6 — Solution

Here A_i follows the order of alternatives in the exam: Single-origin, Balanced, Classic. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly.

First, $0.45 + 0.35 + 0.20 = 1$. Then

$$EMV(A_1) = (0.45)(86) + (0.35)(44) + (0.20)(-12) = 38.7 + 15.4 - 2.4 = 51.7,$$
$$EMV(A_2) = (0.45)(64) + (0.35)(58) + (0.20)(22) = 28.8 + 20.3 + 4.4 = 53.5,$$
$$EMV(A_3) = (0.45)(42) + (0.35)(47) + (0.20)(37) = 18.9 + 16.45 + 7.4 = 42.75.$$

The highest EMV is $\max\{51.7, 53.5, 42.75\} = 53.5$. Therefore, choose A_2, the balanced product mix, with expected profit \$53,500.

5. Aquarium Program Scale **[C6-2]**

An aquarium is deciding the scale of a temporary education program. Payoffs are in thousands of dollars.

		High	Expected	Low
A_i	$P(S_j)$	0.30	0.40	0.30
Major		98	38	-24
Standard		73	53	18
Compact		52	47	34
Regular tours only		27	27	27

What decision would be selected using the EMV or Bayes criterion? Justify your answer.

C6 — Solution

Here A_i follows the order of alternatives in the exam: Major, Standard, Compact, Regular tours only. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly.

The probabilities are valid because $0.30 + 0.40 + 0.30 = 1$.

$$EMV(A_1) = (0.30)(98) + (0.40)(38) + (0.30)(-24) = 29.4 + 15.2 - 7.2 = 37.40,$$
$$EMV(A_3) = (0.30)(52) + (0.40)(47) + (0.30)(34) = 15.6 + 18.8 + 10.2 = 44.60,$$
$$EMV(A_2) = (0.30)(73) + (0.40)(53) + (0.30)(18) = 21.9 + 21.2 + 5.4 = 48.50,$$
$$EMV(A_4) = (0.30)(27) + (0.40)(27) + (0.30)(27) = 27.00.$$

Because 48.50 is the highest EMV, the Bayes decision is A_2, the standard program, with expected payoff \$48,500.

6. Wind-Farm Maintenance Staffing **[C6-2]**

A wind-farm operator is choosing a maintenance staffing plan. Payoffs are in thousands of dollars.

		Surge	Strong	Steady	Slow
A_i	$P(S_j)$	0.20	0.30	0.35	0.15
Rapid hiring		105	64	24	-32
Moderate hiring		79	68	48	12
Current staff		53	55	51	42

What decision would be selected using the EMV or Bayes criterion? Justify your answer.

C6 — Solution

Here A_i follows the order of alternatives in the exam: Rapid hiring, Moderate hiring, Current staff. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly.

The probability check is $0.20 + 0.30 + 0.35 + 0.15 = 1$.

$$EMV(A_2) = (0.20)(79) + (0.30)(68) + (0.35)(48) + (0.15)(12) = 15.8 + 20.4 + 16.8 + 1.8 = 54.8,$$
$$EMV(A_1) = (0.20)(105) + (0.30)(64) + (0.35)(24) + (0.15)(-32) = 21 + 19.2 + 8.4 - 4.8 = 43.8,$$
$$EMV(A_3) = (0.20)(53) + (0.30)(55) + (0.35)(51) + (0.15)(42) = 10.6 + 16.5 + 17.85 + 6.3 = 51.25.$$

Thus $\max\{43.8, 54.8, 51.25\} = 54.8$, so choose A_2, moderate hiring, with expected payoff \$54,800.

7. Concert Merchandise Order **[C7-2]**

A concert vendor must choose between a large and a small merchandise order. Payoffs are in thousands of dollars.

A_i	High (S_1)	Low (S_2)
Large	80	20
Small	50	40

Let $P(S_1) = p$. State the preferred alternative for every probability value ($0 \le p \le 1$).

C7 — Solution

Here A_i follows the order of alternatives in the exam: Large, Small. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly. The listed states are assumed mutually exclusive and exhaustive, so their probabilities sum to one.

$$EMV(A_1) = 80p + 20(1-p) = 20 + 60p,$$
$$EMV(A_2) = 50p + 40(1-p) = 40 + 10p.$$

At the threshold, $20 + 60p = 40 + 10p$, so $50p = 20$ and $p = \frac{2}{5} = 0.4$. Since A_2 has the higher EMV at $p = 0$ and A_1 has the higher EMV at $p = 1$,

$$\begin{cases} A_2, & 0 \le p < \frac{2}{5}, \\ A_1 \text{ and } A_2 \text{ are tied}, & p = \frac{2}{5}, \\ A_1, & \frac{2}{5} < p \le 1. \end{cases}$$



8. Meal-Kit Launch **[C7-2]**

A meal-kit company is comparing a bold launch with a measured launch. Payoffs are in thousands of dollars.

	High	Moderate	Low
A_i	S_1	S_2	S_3
Bold	60	30	0
Measured	35	45	30

Suppose $P(S_1) = p$, $P(S_2) = \frac{p}{3}$. State the preferred alternative for every feasible probability value of p.

C7 — Solution

Here A_i follows the order of alternatives in the exam: Bold, Measured. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly. The listed states are assumed mutually exclusive and exhaustive, so their probabilities sum to one.

Because the three state probabilities must sum to one,

$$P(S_3) = 1 - p - \frac{p}{3} = 1 - \frac{4p}{3}.$$

Nonnegative probabilities require $p \geq 0$ and $1 - \frac{4p}{3} \geq 0$, so the feasible range is $0 \leq p \leq \frac{3}{4}$.

$$EMV(A_1) = 60p + 30\left(\frac{p}{3}\right) + 0\left(1 - \frac{4p}{3}\right) = 70p,$$

$$EMV(A_2) = 35p + 45\left(\frac{p}{3}\right) + 30\left(1 - \frac{4p}{3}\right) = 30 + 10p.$$

Equating the EMVs gives $70p = 30 + 10p$, hence

$$60p = 30 \quad \Rightarrow \quad p = \frac{1}{2}.$$

Therefore,

$$\begin{cases} A_2, & 0 \leq p < \frac{1}{2}, \\ A_1 \text{ and } A_2 \text{ are tied}, & p = \frac{1}{2}, \\ A_1, & \frac{1}{2} < p \leq \frac{3}{4}. \end{cases}$$

9. Publishing-Platform Bid **[C7-2]**

A publishing firm must choose between a standard platform bid and an experimental platform bid. Payoffs are in thousands of dollars.

	Immediate Revision		Delayed
A_i	S_1	S_2	S_3
Standard	30	70	70
Experimental	50	50	30

Suppose $P(S_1) = 3p$, $P(S_2) = p$. State the preferred alternative for every feasible probability value of p.

C7 — Solution

Here A_i follows the order of alternatives in the exam: Standard, Experimental. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly. The listed states are assumed mutually exclusive and exhaustive, so their probabilities sum to one.

The remaining probability is

$$P(S_3) = 1 - 3p - p = 1 - 4p.$$

The probability conditions $p \geq 0$ and $1 - 4p \geq 0$ give the feasible range $0 \leq p \leq \frac{1}{4}$.

$$EMV(A_1) = 30(3p) + 70p + 70(1 - 4p) = 70 - 120p,$$
$$EMV(A_2) = 50(3p) + 50p + 30(1 - 4p) = 30 + 80p.$$

At indifference,

$$70 - 120p = 30 + 80p \quad \Rightarrow \quad 40 = 200p \quad \Rightarrow \quad p = \frac{1}{5} = 0.2.$$

Here A_1 is preferred below the threshold, while the faster-growing EMV of A_2 makes it preferred above the threshold:

$$\begin{cases} A_1, & 0 \leq p < \frac{1}{5}, \\ A_1 \text{ and } A_2 \text{ are tied}, & p = \frac{1}{5}, \\ A_2, & \frac{1}{5} < p \leq \frac{1}{4}. \end{cases}$$



10 Summarized Solutions

10.1 Artisan Apiary Expansion C5–2

The best-payoff vector is $(76, 49, 35)$. The regret table and row maxima are

	S_1	S_2	S_3	$\max r$
A_1	0	15	49	49
A_2	18	0	12	18
A_3	38	6	0	38

Thus $\min\{49, 18, 38\} = 18$, so select A_2, the medium expansion.

10.2 Regional Theater Capacity C5–2

The best-payoff vector is $(108, 61, 34)$. The regret table and row maxima are

	S_1	S_2	S_3	$\max r$
A_1	0	21	62	62
A_2	26	0	20	26
A_3	49	9	0	49
A_4	74	27	0	74

Thus $\min\{62, 26, 49, 74\} = 26$, so select A_2, add one seating section.

10.3 Emergency Shelter Staffing C5–2

The best-payoff vector is $(96, 64, 49, 38)$. The regret table and row maxima are

	S_1	S_2	S_3	S_4	$\max r$
A_1	0	10	25	62	62
A_2	22	0	0	26	26
A_3	48	13	3	0	48

Thus $\min\{62, 26, 48\} = 26$, so select A_2, flexible staffing.

10.4 Coffee Roastery Product Mix C6–2

Since $0.45 + 0.35 + 0.20 = 1$,

$$EMV(A_1) = (0.45)(86) + (0.35)(44) + (0.20)(-12) = 51.7,$$
$$EMV(A_2) = (0.45)(64) + (0.35)(58) + (0.20)(22) = 53.5,$$
$$EMV(A_3) = (0.45)(42) + (0.35)(47) + (0.20)(37) = 42.75.$$

The largest EMV is 53.5, so select A_2, the balanced product mix (\$53,500).

10.5 Aquarium Program Scale C6–2

$$EMV(A_1) = 37.40, \qquad EMV(A_3) = 44.60,$$
$$EMV(A_2) = 48.50, \qquad EMV(A_4) = 27.00.$$

Using probabilities $(0.30, 0.40, 0.30)$, the largest EMV is 48.50, so select A_2, the standard program (\$48,500).

10.6 Wind-Farm Maintenance Staffing C6–2

$$EMV(A_2) = (0.20)(79) + (0.30)(68) + (0.35)(48) + (0.15)(12) = 54.8,$$
$$EMV(A_1) = (0.20)(105) + (0.30)(64) + (0.35)(24) + (0.15)(-32) = 43.8,$$
$$EMV(A_3) = (0.20)(53) + (0.30)(55) + (0.35)(51) + (0.15)(42) = 51.25.$$

The largest EMV is 54.8, so select A_2, moderate hiring (\$54,800).

10.7 Concert Merchandise Order C7–2

$$EMV(A_1) = 20 + 60p, \qquad EMV(A_2) = 40 + 10p.$$

Indifference requires $20 + 60p = 40 + 10p$, so $p = \frac{2}{5} = 0.4$. Therefore

$$A_2 \ (0 \le p < \tfrac{2}{5}); \quad A_1 = A_2 \ (p = \tfrac{2}{5}); \quad A_1 \ (\tfrac{2}{5} < p \le 1).$$

10.8 Meal-Kit Launch C7–2

Here $P(S_3) = 1 - \frac{4p}{3}$, so $0 \le p \le \frac{3}{4}$. Also,

$$EMV(A_1) = 70p, \qquad EMV(A_2) = 30 + 10p.$$

Indifference requires $70p = 30 + 10p$, so $p = \frac{1}{2}$. Therefore

$$A_2 \ (0 \le p < \tfrac{1}{2}); \quad A_1 = A_2 \ (p = \tfrac{1}{2}); \quad A_1 \ (\tfrac{1}{2} < p \le \tfrac{3}{4}).$$

10.9 Publishing-Platform Bid C7–2

Here $P(S_3) = 1 - 4p$, so $0 \le p \le \frac{1}{4}$. Also,

$$EMV(A_1) = 70 - 120p, \qquad EMV(A_2) = 30 + 80p.$$

Indifference requires $70 - 120p = 30 + 80p$, so $p = \frac{1}{5} = 0.2$. Therefore

$$A_1 \ (0 \le p < \tfrac{1}{5}); \quad A_1 = A_2 \ (p = \tfrac{1}{5}); \quad A_2 \ (\tfrac{1}{5} < p \le \tfrac{1}{4}).$$